

# DS4420 Final Project Bayesian Model

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### Objective

We created a Bayesian predictive model that predicts the likelihood of pet adoption based on the animal's identified key traits.

### Setup

#### Read Data

```
pets <- read.csv("pet_adoption_data.csv")
head(pets)
```

```
##   PetID PetType      Breed AgeMonths  Color  Size  WeightKg Vaccinated
## 1   500   Bird    Parakeet     131 Orange  Large   5.039768         1
## 2   501  Rabbit     Rabbit      73  White  Large  16.086727         0
## 3   502   Dog Golden Retriever    136 Orange Medium  2.076286         0
## 4   503   Bird    Parakeet      97  White  Small   3.339423         0
## 5   504  Rabbit     Rabbit     123  Gray  Large  20.498100         0
## 6   505   Dog    Labrador      70  Brown  Large  20.986261         0
##   HealthCondition TimeInShelterDays AdoptionFee PreviousOwner
## 1                0                27          140            0
## 2                0                 8          235            0
## 3                0                85          385            0
## 4                0                61          217            1
## 5                0                28           14            1
## 6                0                87          301            1
##   AdoptionLikelihood
## 1                    0
## 2                    0
## 3                    0
## 4                    0
## 5                    0
## 6                    0
```

## Quick Data Checks

```
table(pets$AdoptionLikelihood, useNA = "ifany")
```

```
##
##      0      1
## 1348   659
```

```
model_cols <- c(
  "AdoptionLikelihood", "AgeMonths", "Breed", "PetType", "HealthCondition",
  "Vaccinated", "AdoptionFee", "TimeInShelterDays", "PreviousOwner", "WeightKg"
)
missing_cols <- setdiff(model_cols, names(pets))
if (length(missing_cols) > 0) stop("Missing columns: ", paste(missing_cols, collapse = ", "))
colSums(is.na(pets[model_cols]))
```

```
## AdoptionLikelihood      AgeMonths      Breed      PetType
##              0              0              0              0
##      HealthCondition      Vaccinated      AdoptionFee      TimeInShelterDays
##              0              0              0              0
##      PreviousOwner      WeightKg
##              0              0
```

## Preprocess

```
pets_scaled <- pets
pets_scaled$Breed <- fct_lump(as.factor(pets$Breed), n = 5)

# numeric features get scaled
pets_scaled$AgeMonths <- as.numeric(scale(pets$AgeMonths))
pets_scaled$AdoptionFee <- as.numeric(scale(pets$AdoptionFee))
pets_scaled$WeightKg <- as.numeric(scale(pets$WeightKg))
pets_scaled$TimeInShelterDays <- as.numeric(scale(pets$TimeInShelterDays))

# categorical features
pets_scaled$PetType <- as.factor(pets$PetType)
pets_scaled$HealthCondition <- as.factor(pets$HealthCondition)
pets_scaled$Vaccinated <- as.factor(pets$Vaccinated)
pets_scaled$PreviousOwner <- as.factor(pets$PreviousOwner)

# check Bernoulli outcome
stopifnot(all(pets_scaled$AdoptionLikelihood %in% c(0, 1)))

n <- nrow(pets_scaled)
train_idx <- sample(1:n, size = 0.8 * n)

train_data <- pets_scaled[train_idx, ]
test_data <- pets_scaled[-train_idx, ]
```

## Fit Bayes Model

```
n_chains <- 4
n_iter <- 2000
n_warmup <- 1000

# stronger priors might help..?
prior <- c(
  prior(normal(0, 0.5), class = "b"),
  prior(normal(0, 1), class = "Intercept")
)

# try taking out breed
model <- brm(
  AdoptionLikelihood ~ AgeMonths + PetType + HealthCondition + Vaccinated +
    AdoptionFee + TimeInShelterDays + PreviousOwner + WeightKg + (1 | Breed)
  ,
  family = bernoulli("logit"),
  data = train_data,
  chains = n_chains,
  cores = 4,
  seed = 42,
  warmup = n_warmup,
  iter = n_iter,
  refresh = 100,
  control = list(
    adapt_delta = 0.99,
    max_treedepth = 15
  )
)
```

```
## Compiling Stan program...
```

```
## WARNING: Rtools is required to build R packages, but is not currently installed.
```

```
##
```

```
## Please download and install the appropriate version of Rtools for 4.3.1 from
```

```
## https://cran.r-project.org/bin/windows/Rtools/.
```

```
## Trying to compile a simple C file
```

```
## Running "C:/PROGRA~1/R/R-43~1.1/bin/x64/Rcmd.exe" SHLIB foo.c
```

```
## using C compiler: 'gcc.exe (GCC) 12.3.0'
```

```
## gcc -I"C:/PROGRA~1/R/R-43~1.1/include" -DNDEBUG -I"C:/Users/jjams/AppData/Local/R/win-library/4.3
```

```
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
```

```
## In file included from C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/Core:19
```

```
## from C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/Dense:1
```

```
## from C:/Users/jjams/AppData/Local/R/win-library/4.3/StanHeaders/include/stan/math/p
```

```
## from <command-line>:
```

```
## C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10
```

```
## 679 | #include <cmath>
```

```
## | ~~~~~
```

```
## compilation terminated.
```

```
## make: *** [C:/PROGRA~1/R/R-43~1.1/etc/x64/Makeconf:265: foo.o] Error 1
```

```
## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.1 from
## https://cran.r-project.org/bin/windows/Rtools/.
```

```
## Start sampling
```

```
saveRDS(model, "bayes_model_fit_5.rds")
```

## Load Saved Model

```
model <- readRDS("bayes_model_fit_5.rds")
```

## Outcomes

```
library("pROC")
summary(model)
```

```
## Family: bernoulli
## Links: mu = logit
## Formula: AdoptionLikelihood ~ AgeMonths + PetType + HealthCondition + Vaccinated + AdoptionFee + TimeInShelterDays
## Data: train_data (Number of observations: 1605)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~Breed (Number of levels: 6)
## Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)      1.68      1.02      0.58      4.50 1.00      1247      1510
##
## Regression Coefficients:
## Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -2.25      1.84     -5.94      1.48 1.00      1318      1300
## AgeMonths           -0.57      0.07     -0.70     -0.44 1.00      6084      2988
## PetTypeCat          -0.13      2.33     -4.83      4.51 1.00      1395      1453
## PetTypeDog           1.25      2.38     -3.40      6.18 1.00      1562      1433
## PetTypeRabbit       -0.12      2.72     -5.82      5.41 1.00      1779      1567
## HealthCondition1    -2.20      0.22     -2.66     -1.79 1.00      6494      2744
## Vaccinated1         2.09      0.18      1.75      2.44 1.00      6299      2878
## AdoptionFee        -0.07      0.06     -0.19      0.05 1.00      6900      2845
## TimeInShelterDays   0.06      0.06     -0.06      0.19 1.00      6115      2814
## PreviousOwner1     -0.02      0.14     -0.29      0.25 1.00      6005      2728
## WeightKg            0.02      0.06     -0.10      0.15 1.00      6617      2579
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
pred_probs <- fitted(model, newdata = test_data)[, "Estimate"]
pred_class <- ifelse(pred_probs > 0.5, 1, 0)

y_test <- test_data$AdoptionLikelihood
mean(pred_class == y_test)
```

```
## [1] 0.7512438
```

```
roc_obj <- roc(y_test, pred_probs)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
auc(roc_obj)
```

```
## Area under the curve: 0.8341
```